

Grade 5 lessons

Fractions, decimals and order of operations

IN THIS BOOK

1. Adding and subtracting fractions
2. Multiplying fractions
3. Decimal operations
4. Order of operations
5. Volume and coordinates

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 5 Kahoot and Blooket sets, and one stop on the Grade 5 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Adding and subtracting fractions

THE BIG IDEA

Fractions must have the same bottom number before you can add or subtract them.

HOW TO DO IT

1. Find a bottom number that both bottoms divide into. The smallest one is best.
2. Rewrite each fraction with that bottom number.
3. Add or subtract the tops. Keep the bottom the same.
4. Simplify the answer if you can.

WORKED EXAMPLES

$$\frac{1}{2} + \frac{1}{3} = ?$$

Use sixths. $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$. Add the tops: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

$$\frac{3}{4} - \frac{1}{6} = ?$$

Use twelfths. $\frac{3}{4} = \frac{9}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Subtract: $\frac{9}{12} - \frac{2}{12} = \frac{7}{12}$.

WATCH OUT

Never add the bottoms. $\frac{1}{2} + \frac{1}{3}$ is not $\frac{2}{5}$.

WORDS TO KNOW

common denominator a bottom number that both fractions can use

Multiplying fractions

THE BIG IDEA

To multiply fractions, multiply the tops and multiply the bottoms. The word "of" means multiply.

HOW TO DO IT

1. Multiply the top numbers.
2. Multiply the bottom numbers.
3. Simplify the answer.
4. For a whole number, put it over 1. 4 is the same as $\frac{4}{1}$.

WORKED EXAMPLES

$$\frac{2}{3} \times \frac{3}{4} = ?$$

Tops: $2 \times 3 = 6$. Bottoms: $3 \times 4 = 12$.

$\frac{6}{12}$ simplifies to $\frac{1}{2}$.

What is $\frac{3}{4}$ of 20?

$20 \div 4 = 5$. That is one quarter. Three quarters is $5 \times 3 = 15$.

WATCH OUT

You do not need a common bottom number to multiply. Multiplying by a fraction less than 1 makes the answer smaller, not bigger.

WORDS TO KNOW

of in fraction problems this means multiply

Decimal operations

THE BIG IDEA

Decimals follow the same rules as whole numbers. The only extra job is keeping the decimal point in the right place.

HOW TO DO IT

1. To add or subtract, line up the decimal points. Fill gaps with zeros.
2. To multiply, ignore the point and multiply. Then count the decimal places in the question and put that many in the answer.
3. To multiply by 10, 100 or 1000, move the point right 1, 2 or 3 places.
4. To divide by 10, or multiply by 0.1, move the point left 1 place.

WORKED EXAMPLES

$3.45 + 2.7 = ?$

Write 2.7 as 2.70. Then $3.45 + 2.70 = 6.15$.

$2.4 \times 3 = ?$

$24 \times 3 = 72$. There is 1 decimal place in the question, so the answer is 7.2.

WATCH OUT

Line up the decimal points, not the last digits.

WORDS TO KNOW

decimal place a position after the decimal point

Order of operations

THE BIG IDEA

When a problem has more than one operation, everyone must do them in the same order so everyone gets the same answer.

HOW TO DO IT

1. Brackets first.
2. Then powers, like 3^2 .
3. Then multiply and divide, working left to right.
4. Then add and subtract, working left to right.

WORKED EXAMPLES

$3 + 4 \times 2 = ?$

Multiply first: $4 \times 2 = 8$. Then add: $3 + 8 = 11$.

$20 - 2 \times 3^2 = ?$

Power first: $3^2 = 9$. Multiply: $2 \times 9 = 18$. Subtract: $20 - 18 = 2$.

WATCH OUT

Do not just work left to right. $3 + 4 \times 2$ is 11, not 14. Use brackets if you want the adding done first: $(3 + 4) \times 2 = 14$.

WORDS TO KNOW

operation

adding, subtracting, multiplying or dividing

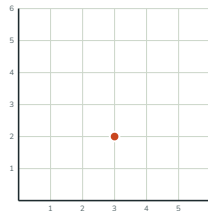
power

a small raised number that says how many times to multiply

Volume and coordinates

THE BIG IDEA

Volume is the space inside a solid shape. A coordinate grid uses two numbers to say exactly where a point is.



The point (3, 2): go 3 across, then 2 up.

HOW TO DO IT

1. Volume of a box = length $\times$ width $\times$ height.
2. Volume is measured in cubic units.
3. A point is written (x, y). The first number is how far across. The second is how far up.
4. If you know the volume and two sides, divide to find the third.

WORKED EXAMPLES

**A box is 4 long, 3 wide and 2 tall.
What is its volume?**

$$4 \times 3 \times 2 = 24 \text{ cubic units.}$$

**A box has volume 60. It is 5 long
and 3 wide. How tall is it?**

$$5 \times 3 = 15. \text{ Then } 60 \div 15 = 4. \text{ It is 4 tall.}$$

WATCH OUT

(3, 2) and (2, 3) are different points. Across always comes first.

WORDS TO KNOW

volume

the space inside a solid

coordinates

the pair of numbers that locate a point